Projection Matrix: A projection matrix P is a square matrix that satisfies condition $P^2 = P$.

If P also Symetric $P^T = P$, it is an Orthogonal Projection. Other wise it is an oblique projection.

The projection matrix onto the column space of matrix A is

$$P = A(A^TA)^{-1}A' \text{ or } \frac{AA^T}{A^TA}$$

Provided A has Columns rank.

Example ①. Projection onto X-axis in R^2 is $P\begin{bmatrix}1&0\\0&0\end{bmatrix}$

Soln: given $P = \begin{bmatrix}1&0\\0&0\end{bmatrix}$

Check $P^2 = \begin{bmatrix}1&0\\0&0\end{bmatrix}\begin{bmatrix}1&0\\0&0\end{bmatrix} = P$.

$\Rightarrow$ P is a projection matrix

Example ②. Projection on the line spanned by vector $u(3,4)$.

Soln: We find the projection matrix onto the line spanned by Vector $u = (3,4)$

i.e. $$P = \frac{UU^T}{U^TU} = \frac{\begin{bmatrix}3\\4\end{bmatrix}(3,4)}{(3,4)\begin{pmatrix}3\\4\end{pmatrix}} = \frac{\begin{bmatrix}9&12\\12&16\end{bmatrix}}{25}$$

$$P = \frac{1}{25}\begin{bmatrix}9&12\\12&16\end{bmatrix}$$

Verify $P^2 = \frac{1}{25}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix} \cdot \frac{1}{25}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$

$= \frac{1}{25}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix} = P.$

$\Rightarrow$ P is a projection Matrix

Example 3: Find the projection onto span of vectors
(1,0,1) (0,1,1).

Sol: The above vector can be written as matrix form. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ we find the projection matrix of A i.e., $P = A(A'A)^{-1}A^T$

$= \begin{bmatrix} 2/3 & -1/3 & 1/3 \\ -1/3 & 2/3 & 1/3 \\ 1/3 & 1/3 & 2/3 \end{bmatrix}$

Verification $P^2 = \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \cdot \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

$= \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} = P.$

$\Rightarrow P^2 = P$, Therefore P is the projection matrix

Ex. 4. Find the projection matrix P onto column space of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$

Soln: By using the projection matrix formula:

$$P = A(A^TA)^{-1}.A^T$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \left[\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \right]^{-1} \cdot \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Verification $P^2 = \frac{1}{3} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

$$= \frac{1}{3} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Ex: 5. Find the projection matrix on the line spanned by a vector (1,1,1).

Soln: Let the given vector ~~(1,1,1)~~ $u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Hence the projection Matrix $P = \frac{UU^T}{U^T.U}$

$$= \frac{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}}{\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

P is the required projection matrix onto the line spanned by vector (1,1,1).

Verification: left to the student.

Ex. 6. Find the projection matrix onto the plane $x+y+z=0$ in $\mathbb{R}^3$.

Soln: The plane is given by

$$x+y+z=0$$

its normal vector is $n = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Formula for projection onto a plane i.e

The projection matrix onto a plane with normal

Vector $P = I - \frac{n \cdot n^T}{n^T \cdot n}$

Now $n^T \cdot n = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1^2+1^2+1^2 = 3$

$$n \cdot n^T = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

$\therefore$ The projection matrix

$$P = I - \frac{n n^T}{n^T \cdot n} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Verification (Idempotence) is

$$P^2 = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = P.$$

$\Rightarrow P$ is the required projection matrix

HW:

Ex. 7. Find the projection Matrix P onto the plane $2x+y+z=3$ in $\mathbb{R}^3$.